Anik Burman

Doctoral Candidate in Biostatistics

Johns Hopkins Bloomberg School of Public Health

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RESEARCH INTERESTS

Statistical and causal machine learning under dependence and distribution shift, with applications to spatial, spatio-temporal, environmental, and health data. Methodological interests include semiparametric and doubly robust estimation, transportability, spatial statistics, and distribution-free uncertainty quantification.

EDUCATION

PhD in Biostatistics, Johns Hopkins Bloomberg School of Public Health — 2022–2027 (expected)
Advisors: Abhirup Datta and Elizabeth L. Ogburn
Dissertation: *Causal inference for spatially and temporally dependent data, with applications in environmental health*

Master of Statistics (MStat), Indian Statistical Institute, Kolkata — 2020–2022

Bachelor of Statistics (BStat), Honours, Indian Statistical Institute, Kolkata — 2017–2020

PUBLICATIONS

Accepted and Published

1. Rupam Bhattacharyya, **Anik Burman**, Kalpana Singh, Sayantan Banerjee, Subha Maity, Arnab Auddy, Sarit Kumar Rout, Supriya Lahoti, Rajmohan Panda, and Veerabhadran Baladandayuthapani (2022). Role of multiresolution vulnerability indices in coronavirus disease 2019 (COVID-19) spread in India: A Bayesian model-based analysis. *BMJ Open*, 12(11), e056292. DOI: 10.1136/bmjopen-2021-056292.
2. Lauren N. Deanes, . . . , **Anik Burman**, Abhirup Datta, and Christopher D. Heaney (2025). Relation of wind direction and coal terminal activity patterns with air pollution burden in a community bordering a coal export terminal, Curtis Bay, Maryland, United States of America (USA). *Air Quality, Atmosphere & Health*, 18(9), 2805–2821. DOI: 10.1007/s11869-025-01773-w.

Preprints Under Revision

1. **Anik Burman**, Elizabeth L. Ogburn, and Abhirup Datta (2025). Robust spatial confounding adjustment via basis voting. arXiv Link [Under major revision at the Journal of the American Statistical Association (JASA) Theory & Methods].
2. **Anik Burman**, Sayantan Choudhury, and Debangan Dey (2026). Doubly-unlinked regression for dependent data. arXiv Link. [Under Revision at the Journal of the Royal Statistical Society Series B: Statistical Methodology (JRSSB)].

In Preparation

1. **Anik Burman**, Margaret Gamalo, Promit Ghosal, and Prosenjit Kundu (2026). Transporting randomized trial effects to real-world populations via Riesz-calibrated optimal transport. (arXiv Link)
2. **Anik Burman**, Elizabeth L. Ogburn, Abhirup Datta, Christopher D. Heaney et.al. Causal machine learning for shared environmental exposures: Coal terminal activity and air pollution in Baltimore.
3. **Anik Burman**, Elizabeth L. Ogburn, and Abhirup Datta. Selection-free inference for mode-based estimators with invalid instruments.
4. **Anik Burman**, Elizabeth L. Ogburn, and Abhirup Datta. Causal inference for shift interventions on continuous spatial processes under interference.
5. Anirban Nath, **Anik Burman**, and Debangan Dey. Conformal prediction for spatial processes under inhomogeneous sampling.

Other Preprints

1. **Anik Burman** and Sayantan Banerjee (2021). High-dimensional portfolio optimization using joint shrinkage. (arXiv Link).
2. **Anik Burman**, Sayak Chatterjee, Pramit Ghosh, and Indranil Mukhopadhyay (2021). An artificial intelligence (AI)-enabled agent-based simulation platform for studying COVID-19 pandemic. (arXiv Link).

RESEARCH AND INTERNSHIP EXPERIENCE

PhD Statistics Intern, Pfizer, Cambridge, Massachusetts (MA) 2026
Transporting randomized trial effects to real-world target populations with optimal transport
Supervisor: Prosenjit Kundu

Research Assistant, Indian Institute of Management Indore, India 2022
Shrinkage estimation of precision matrices for high-dimensional portfolio optimization
Supervisor: Sayantan Banerjee

Summer Intern, Centre for Infrastructure, Sustainable Transportation and Urban Planning (CiSTUP), Indian Institute of Science, Bangalore, India 2021
Gender differences and the impact of commute on the daily time use of workers in Indian metropolitan cities. Report
Supervisor: Abdul Rawoof Pinjari

Research Intern, University of Michigan, Ann Arbor, Michigan (MI) 2021
Bayesian vulnerability indices for COVID-19 spread across districts of Odisha, India
Supervisor: Veerabhadran Baladandayuthapani

Big Data Summer Institute, University of Michigan School of Public Health, Ann Arbor, Michigan (MI) 2020
Vaccine trust in incentivized versus non-incentivized states; topic modeling with synonyms. Slides
Supervisor: Johann Gagnon-Bartsch

Summer Research Fellow, Summer Research Fellowship Programme (SRFP), National Institute of Biomedical Genomics (NIBMG), India — 2019
One of 102 fellows selected nationwide in Mathematics and Statistics
Supervisor: Samsiddhi Bhattacharjee

TALKS AND PRESENTATIONS

Robust Spatial Confounding Adjustment via Basis Voting

- American Statistical Association (ASA) Section on Statistical Learning and Data Science (SLDS) Conference, New York, New York (NY) *(invited talk)* — Nov 2026 (upcoming)
- Biostatistics Department Retreat, Johns Hopkins Bloomberg School of Public Health — Oct 2026
- Joint Statistical Meetings (JSM), Boston, Massachusetts (MA) *(invited talk)* — Aug 2026
- Eastern North American Region (ENAR) Spring Meeting, Indianapolis, Indiana (IN) — Mar 2026
- EnviBayes Workshop, Texas A&M University, College Station, Texas (TX) — 2025

Causal Machine Learning for Shared Environmental Exposures: Coal Terminal Activity and Air Pollution in Baltimore

- 2026 Workshop of the American Statistical Association (ASA) Section on Statistics and the Environment (ENVR), Houston, Texas (TX) — Oct 2026

HONORS AND AWARDS

ENAR Distinguished Student Paper Award — 2026

Student Travel Award, SLDS Conference — 2026

Student Travel Award, ENVR 2026 Workshop — 2026

June B. Culley Award, Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health, for the best Schoolwide examination paper (PhD proposal defense) — 2026

Travel Award, EnviBayes Workshop — 2025

David Duncan Centennial Scholarship, Johns Hopkins Bloomberg School of Public Health — 2023–2025

TEACHING AND MENTORING

Teaching Assistant and Guest Lecturer, Johns Hopkins Bloomberg School of Public Health

Statistical Theory II–IV, 140.732–734, PhD students — 2026, Terms 2–4

Methods in Biostatistics I, 140.651, master's students — 2026, Term 1

Statistical Theory I–IV, 140.731–734, PhD students — 2023–2025, Terms 1–4

Student Mentoring

Peer mentor, Incoming first-year PhD student Nowell Phelps — 2024

Peer mentor, Incoming first-year PhD student Sam Fansler — 2023

SERVICE AND LEADERSHIP

Journal referee, *Biometrika*, *Biometrics*

Biostatistics Department Representative, Student Assembly, Johns Hopkins Bloomberg School of Public Health